

GE261004 DESIGN THINKING

Unit III – Ideation and Creative Thinking · Presentation 6 of 6

Concept Selection and Evaluation

Converging as a sequence of screens · clustering and dot voting · the three-lens test · the Pugh matrix · traps in selection

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In this presentation

- A sequence of screens
- Converging well
- Clustering and dot voting
- Three quick screens
- The three-lens test
- The Pugh matrix
- Reading the matrix
- Traps in selection

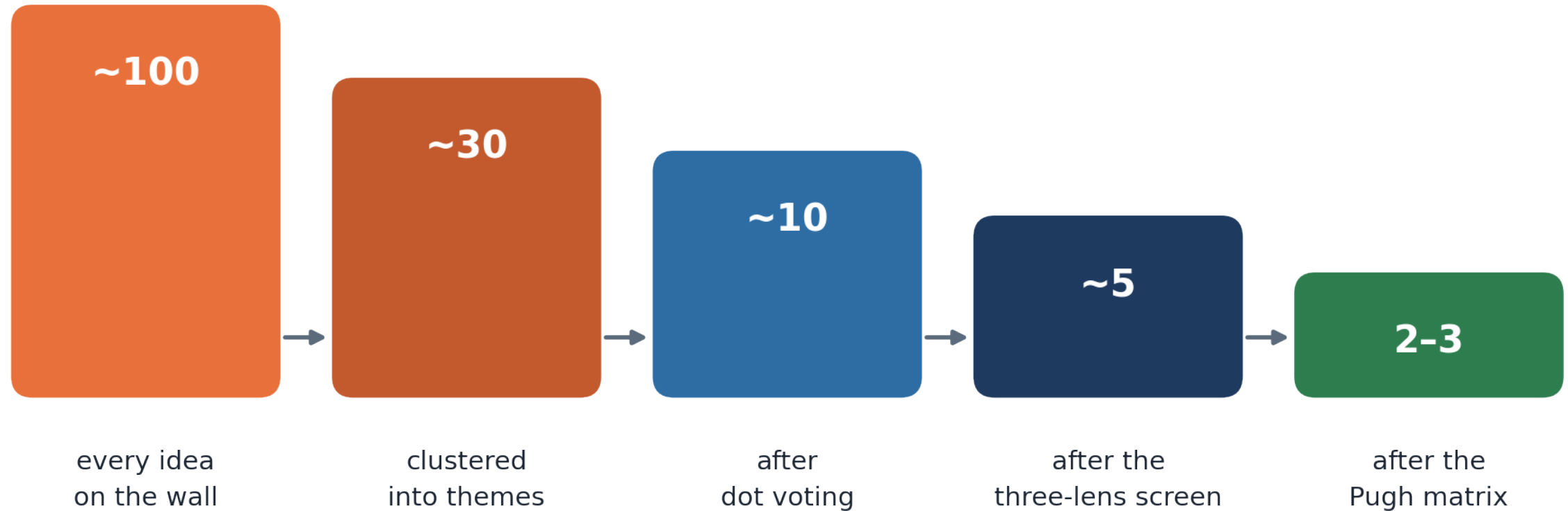


1

From many ideas to a few

Converging is a series of cheap screens, not one long argument

A sequence of screens



keep one wild card through every screen — the safe list is rarely the interesting one

Each screen is quick, and each one answers a different question.

Converging well

- Converge in a separate session from ideating – ideally on a different day
- Agree the criteria before you look at the ideas, or the ideas will choose them
- Screen in stages: cheap and fast first, careful and slow last
- Carry one wild card through every stage, whatever the scores say

KEY POINT Choosing badly is expensive; choosing slowly is not. Spend the time here, before anything gets built.

Clustering and dot voting

- Cluster first: group ideas that are the same idea wearing different words
- Name each cluster in a sentence, so the group is about something
- Then dot vote – three sticky dots each, placed silently and at the same time
- Silence matters: spoken voting turns instantly into anchoring

ACTIVITY Fifteen minutes. Cluster your 108 ideas into themes, name each theme, then vote with three dots each in silence.

Three quick screens

Dot voting

Fast and democratic.
Shows what the room
likes – which is not the
same as what is right.

Four categories

Pick the rational
choice, the most
delightful, the long
shot and the darling.
Forces variety.

Gut check

Which would you be
embarrassed to show
a user? Which would
you be excited to?

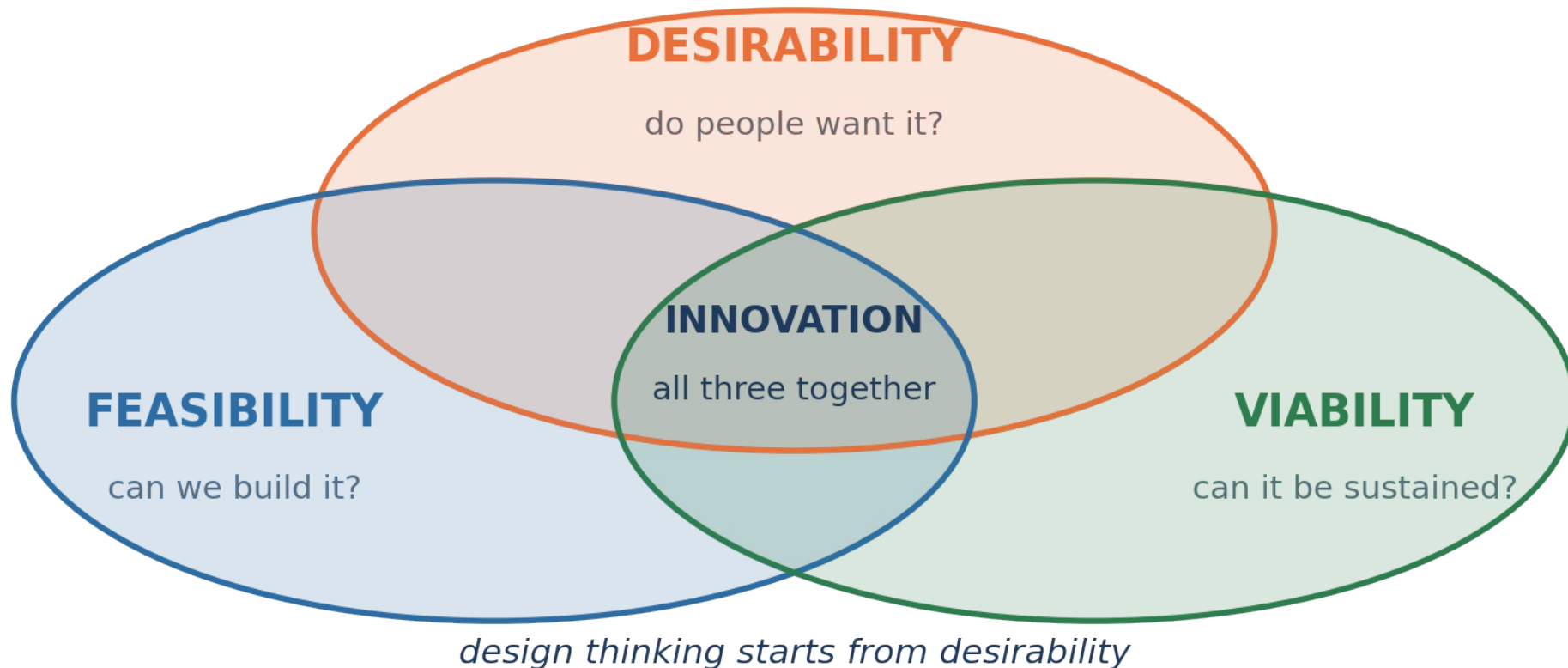
KEY POINT Use at least two screens with different biases. One screen only tells you about the screen.

2

Structured selection

Making the reasons visible, so they can be argued with

The three-lens test



Score each surviving concept on all three – and say which lens it fails.

The Pugh matrix

- Stuart Pugh, Professor of Engineering Design at Strathclyde, set it out in Total Design, 1990
- Pick a datum – usually the existing solution or the most obvious concept
- Score every other concept against the datum on each criterion: better (+), worse (–) or the same (S)
- Count the pluses and minuses; the point is the pattern, not the arithmetic

KEY POINT You are comparing concepts against a baseline, not scoring them out of ten. Relative judgement is far more reliable.

A worked Pugh matrix

Criterion	A: refill points	B: chilled tank	C: bottle service
Cost to install	+	–	S
Ease of upkeep	+	–	–
Queue time	+	S	+
Water quality	S	+	+
Total +	3	1	2
Total –	0	2	1

Datum: the cooler already installed. Each concept is better (+), worse (–) or the same (S).

Reading the matrix

- A concept with no minuses is safe, and often dull – check it is not just the datum again
- A concept with strong pluses and one fatal minus is worth fixing, not dropping
- If everything scores the same, your criteria are not discriminating – rewrite them
- Run it twice with a different datum; a ranking that survives both is real

KEY POINT Use the matrix to find the conversation, not to end it. The interesting column is the one people disagree about.

Traps in selection

What goes wrong

- Criteria invented after the ideas
- The loudest voice weights the scores
- Everything wild is screened out
- One screen, applied once

What to do instead

- Agree criteria first, in writing
- Vote silently and simultaneously
- Protect one wild card by rule
- Two screens with different biases

KEY POINT A selection process that always returns the safe option is not a process. It is a preference with a spreadsheet.

In class: choose your team's concepts

- Cluster, name and dot vote your ideas from the 6-3-5 session
- Take the top five into the three-lens test and say which lens each one risks
- Build a Pugh matrix for the surviving three, with your own criteria
- Leave with two concepts plus one wild card to prototype in Unit IV

ACTIVITY Thirty minutes. Every team pins up its matrix and names the concept it will build first.

THINK ABOUT IT

Your matrix chose the safe concept and your team wanted the wild one. What do you do?

- Were the criteria written before or after the ideas?
- Which criterion is carrying all the weight?
- What would you have to learn to score the wild one properly?
- Could you prototype both this week?

Summary

- Converging is a sequence of cheap screens, run separately from ideating
- Cluster and name first, then dot vote silently
- Use at least two screens with different biases
- The three lenses tell you which way a concept is likely to fail
- Pugh's matrix scores concepts against a datum as +, – or S
- The matrix should start the argument, not finish it

Next: Unit IV – Prototyping and Testing

- Concept visualisation and why we prototype
- Sketching, mock-ups and low-fidelity building
- Rapid prototyping and digital methods
- Testing with users and iterating on what you learn